

SWE 4739

Embedded Software Development

Lecture 8

Hybrid Systems

Sabrina Islam

Lecturer, CSE, IUT

Contact: +8801832239897

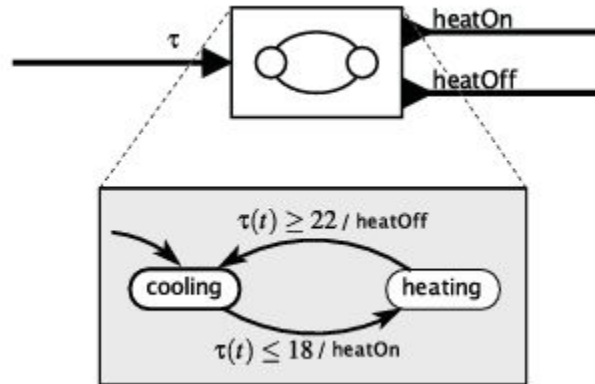
E-mail: sabrinaislam22@iut-dhaka.edu

Actor Model for State Machines

- A state machine can be thought of like an “actor”.
- It behaves like a box that receives inputs and produces outputs

Continuous Inputs

- Instead of getting values at specific moments, we will have : $\tau(t)$
- Thermostat example:
 - Temperature is available at every moment in time
 - Not just at steps like t_1, t_2, t_3

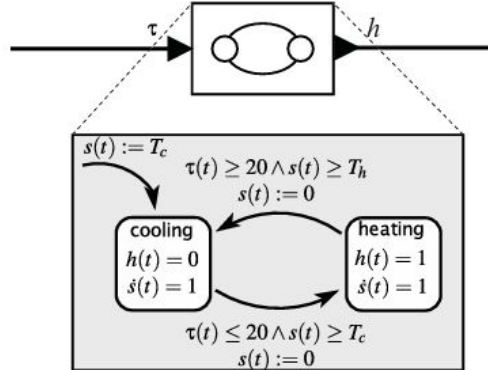


State Refinements

- In a hybrid system, the current state of the state machine has a state refinement that gives the dynamic behavior of the output as a function of the input.

Timed Automata

- A state machine that also keeps track of time
- Each state has a clock: $\dot{s}(t) = 1$
 - The clock increases steadily with time
- Reaction depends on time as well.



Reference:

1. Introduction to Embedded Systems- A Cyber-Physical Systems Approach, Second Edition, E. A. Lee and S. A. Seshia.

Thank You!